Supplementary Materials

Understanding the Molecular Level Interaction of Lithium-ions with Cellulose Based Eutectogel: A Molecular Dynamics Study

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Table S1. System configurations

Component name	Number of the atom presents in the system		
	CEC	EC	MC
LiTFSI	7680	7680	7680
LiDFOB	240	240	240
SN	21600	21600	21600
CEC	6738	-	-
EC	-	6498	-
MC	-	-	5778
Initial box dimension (nm)	58.76 × 39.17 × 19.59	58.34 × 38.90 × 19.45	58.46 × 38.98 × 19.49
Final box dimension (nm)	13.02 × 8.68 × 4.34	12.95 × 8.63 × 4.32	12.88 × 8.59 × 4.29

Table S2. The radial distribution functions and coordination number of the systems

component name	System name	solvation shell (Å)	*CN*	solvation shell (Å)	*CN*
Lithium-ion of LITFSI to the ether moiety of LIDFOB	CEC	2.7	0.03	5.4	0.58
	EC	2.7	0.02	5.4	0.79
	MC	2.7	0.01	5.4	0.87
Lithium-ion of LITFSI to the oxygen moiety of LIDFOB	CEC	3.7	0.04	4.6	0.0
	EC	3.7	0.13	4.6	0.25
	MC	2.8	0.04	4.6	0.26
Lithium-ion of LITFSI to the fluorine moiety of LIDFOB	CEC	2.7	0.18	-	-
	EC	2.7	0.25	-	-
	MC	2.7	0.26	-	-
Lithium-ion of LITFSI to the boron moiety of LIDFOB	CEC	3.8	0.15	5.9	0.32
	EC	3.8	0.33	5.9	0.45
	MC	3.8	0.38	5.9	0.54
Lithium-ion of LITFSI to the nitrogen moiety of LITFSI	CEC	2.6	0.27	4.7	2.13
	EC	2.6	0.86	4.7	2.31
	MC	2.6	0.85	4.7	2.14
Lithium-ion of LITFSI to the oxygen moiety of LITFSI	CEC	2.7	3.20	4.2	5.42
	EC	2.7	3.74	4.2	5.22

	MC	2.7	3.63	4.2	5.18
Lithium-ion of LITFSI to the fluorine moiety of LITFSI	CEC	1.6	0.16	2.6	0.5
	EC	1.6	0.16	2.6	0.5
	MC	1.6	0.16	2.6	0.5
Lithium-ion of LITFSI to the ether moiety of the polymer network	CEC	2.8	0.12	-	-
	EC	2.8	0.13	-	-
	MC	2.8	0.14	-	-
Lithium-ion of LITFSI to the hydroxyl moiety of the polymer network	CEC	2.8	0.01	-	-
	EC	2.8	0.03	-	-
	MC	2.8	0.18	-	-
Lithium-ion of LITFSI to the cyano ethyl moiety of the polymer network	CEC	2.8	0.01	-	-
Lithium-ion of LITFSI to the nitrogen moiety of the SN	CEC	2.8	0.02	6.5	3.90
	EC	2.8	0.04	6.5	3.10
	MC	2.8	0.01	6.5	2.56
Lithium-ion of LIDFOB to the ether moiety of LIDFOB	CEC	2.8	0.16	4.9	0.94
	EC	2.8	0.08	4.9	0.72
	MC	2.8	0.04	4.9	0.78
Lithium-ion of LIDFOB to the oxygen moiety of LIDFOB	CEC	3.6	0.33	5.6	1.04
	EC	3.6	0.12	5.6	0.81

	MC	3.6	0.13	5.6	0.95
Lithium-ion of LIDFOB to the fluorine moiety of LIDFOB	CEC	2.7	0.13	-	-
	EC	2.7	0.25	-	-
	MC	2.7	0.26	-	-
Lithium-ion of LIDFOB to the boron moiety of LIDFOB	CEC	4.0	0.28	-	-
	EC	4.0	0.33	-	-
	MC	4.0	0.30	-	-
Lithium-ion of LIDFOB to the nitrogen moiety of LITFSI	CEC	2.9	0.85	4.8	1.84
	EC	2.9	1.00	4.8	2.32
	MC	2.9	1.01	4.8	2.16
Lithium-ion of LIDFOB to the oxygen moiety of LITFSI	CEC	2.9	3.93	-	-
	EC	2.9	3.80	-	-
	MC	2.9	3.74	-	-
Lithium-ion of LIDFOB to the fluorine moiety of LITFSI	CEC	6.0	14.98	-	-
	EC	6.0	17.36	-	-
	MC	6.0	16.49	-	-
Lithium-ion of LIDFOB to the ether moiety of the polymer network	CEC	3	0.12	4.5	0.21
	EC	3	0.13	4.5	0.23
	MC	3	0.14	4.5	0.28
	CEC	2.5	0.16	-	-

Lithium-ion of LIDFOB to the hydroxyl moiety of the polymer network	EC	4.4	2.08	-	-
	MC	3.9	0.32	-	-
Lithium-ion of LIDFOB to the cyano ethyl moiety of the polymer network	CEC	3.1	0.03	8.2	1.20
Lithium-ion of LIDFOB to the nitrogen moiety of the SN	CEC	3.1	0.08	8.9	12.14
	EC	3.1	0.07	8.9	12.29
	MC	3.1	0.03	8.9	10.09

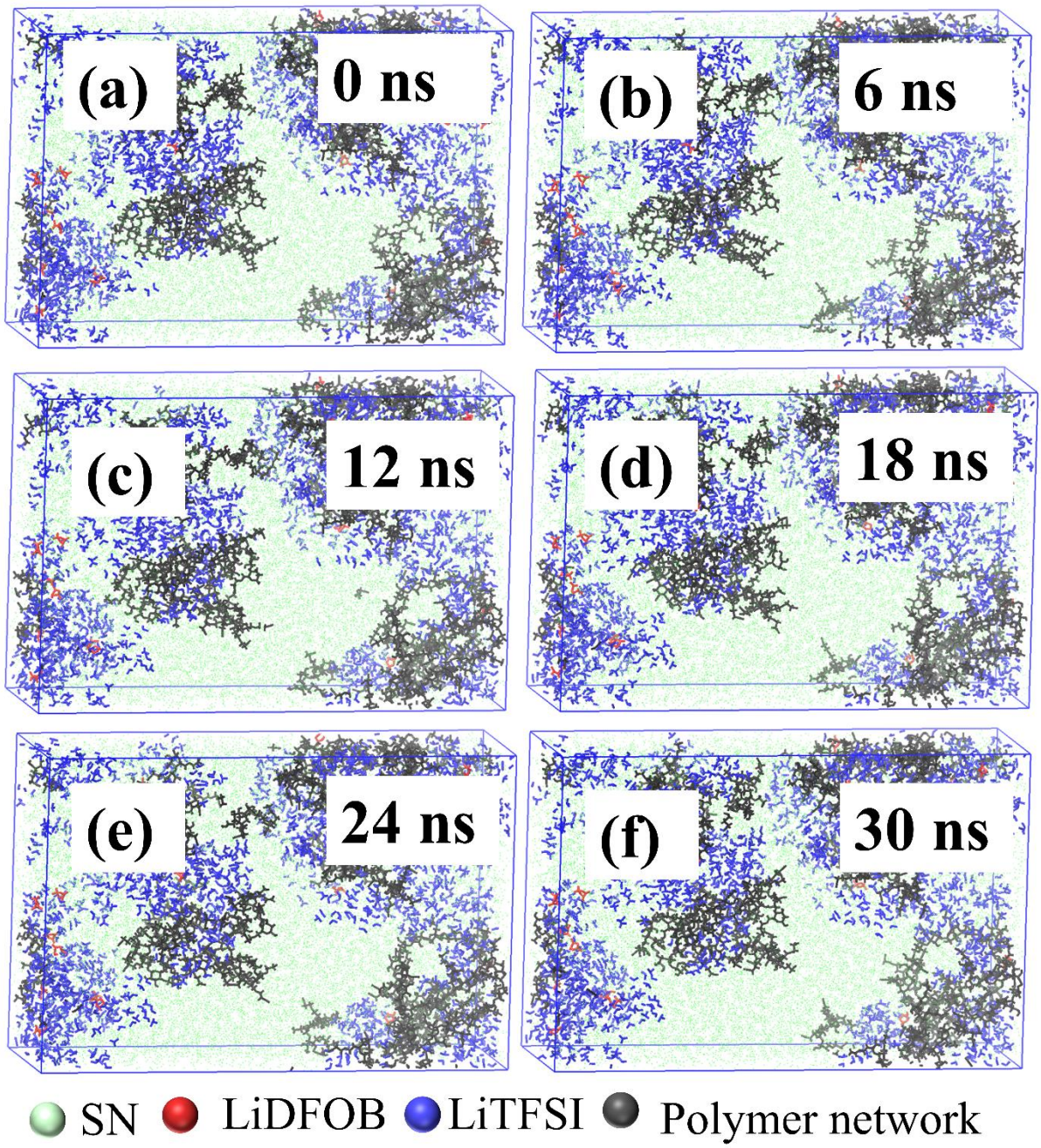


Figure S1. Simulation snapshots for the CEC system at different time interval (a) 0 ns (b) 6ns (c) 12 ns (d) 18 ns (e) 24 ns (f) 30 ns

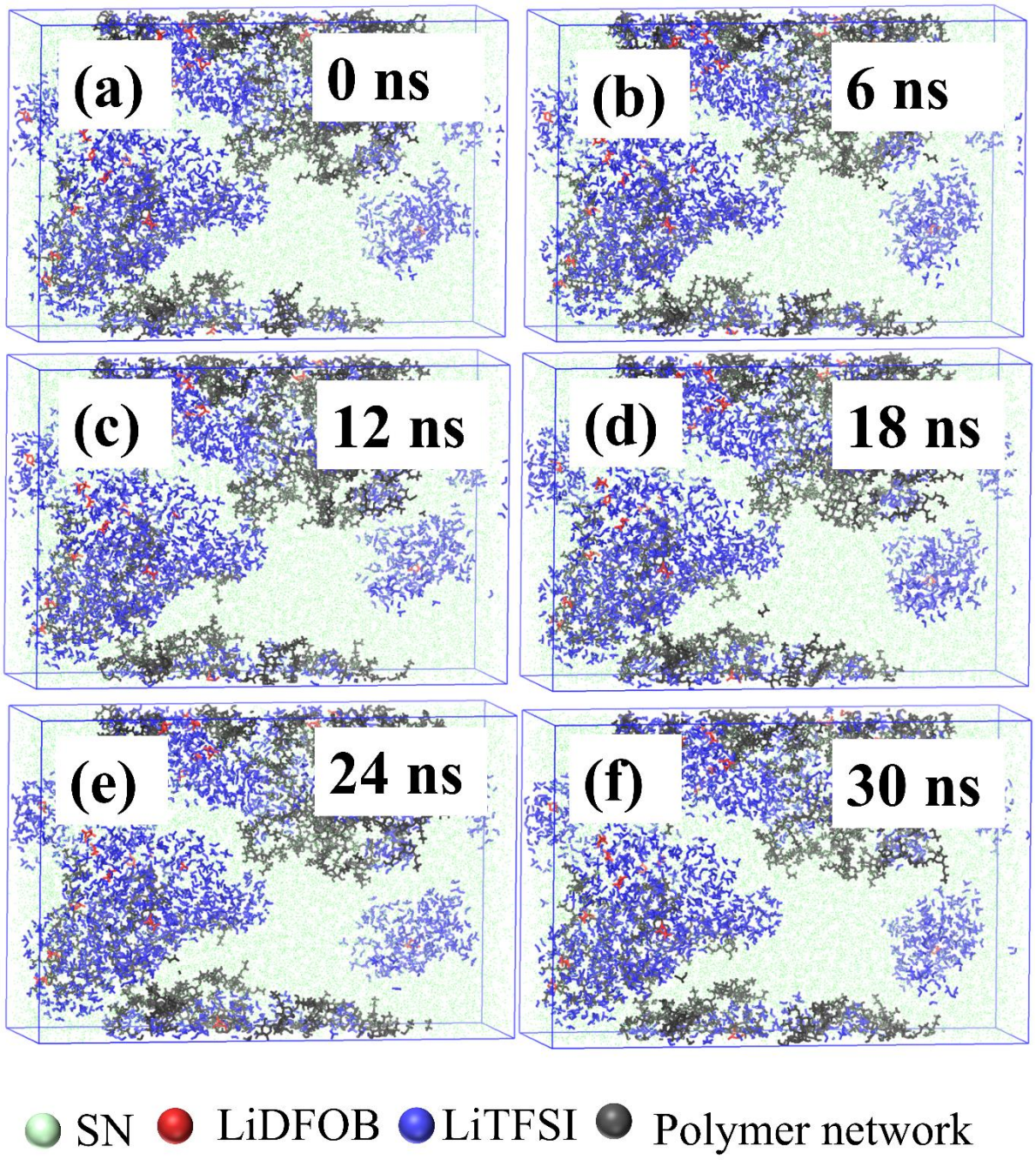


Figure S2. Simulation snapshots for the EC system at different time interval (a) 0 ns (b) 6ns (c) 12 ns (d) 18 ns (e) 24 ns (f) 30 ns

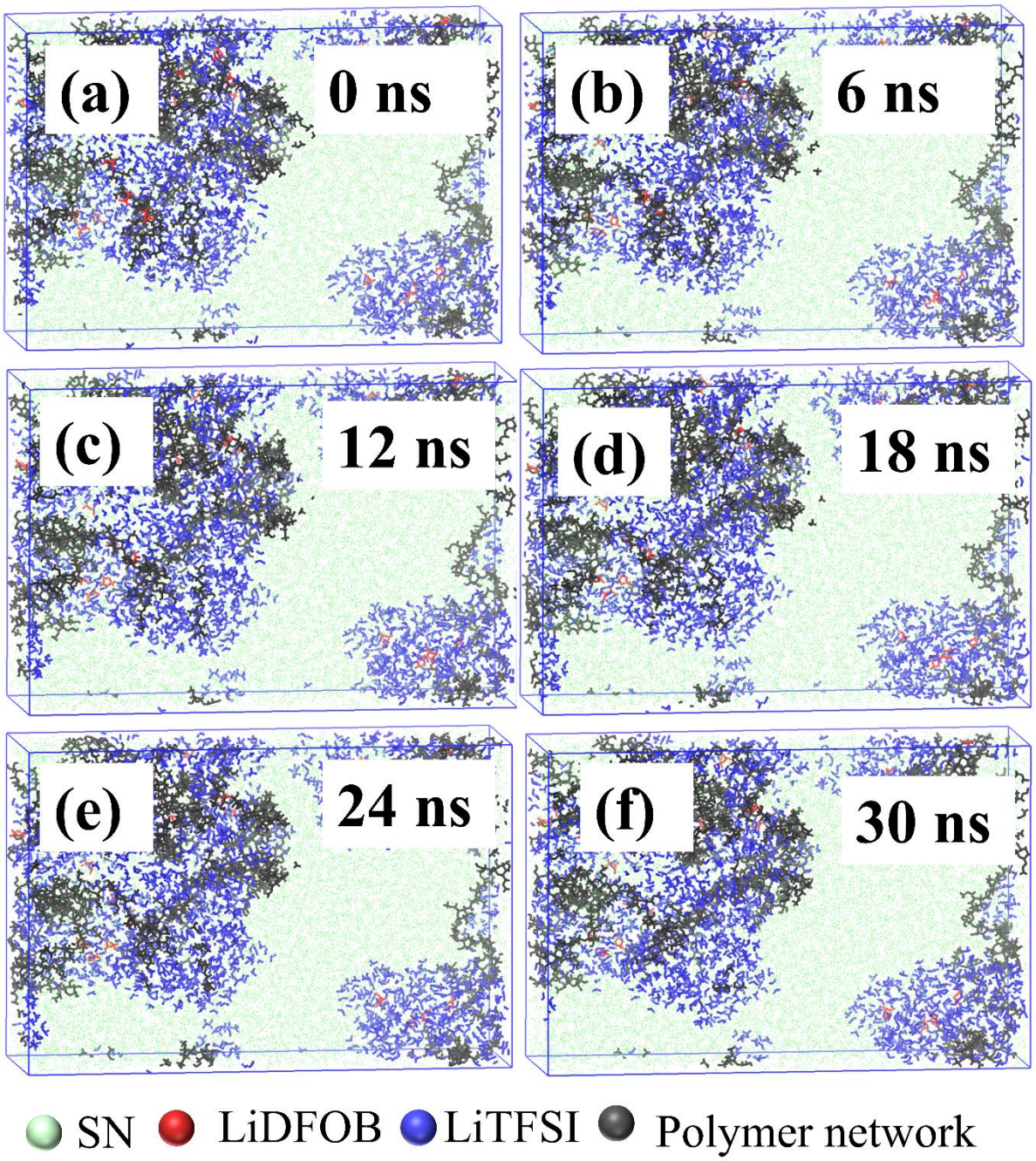


Figure S3. Simulation snapshots for the MC system at different time interval (a) 0 ns (b) 6ns (c) 12 ns (d) 18 ns (e) 24 ns (f) 30 ns

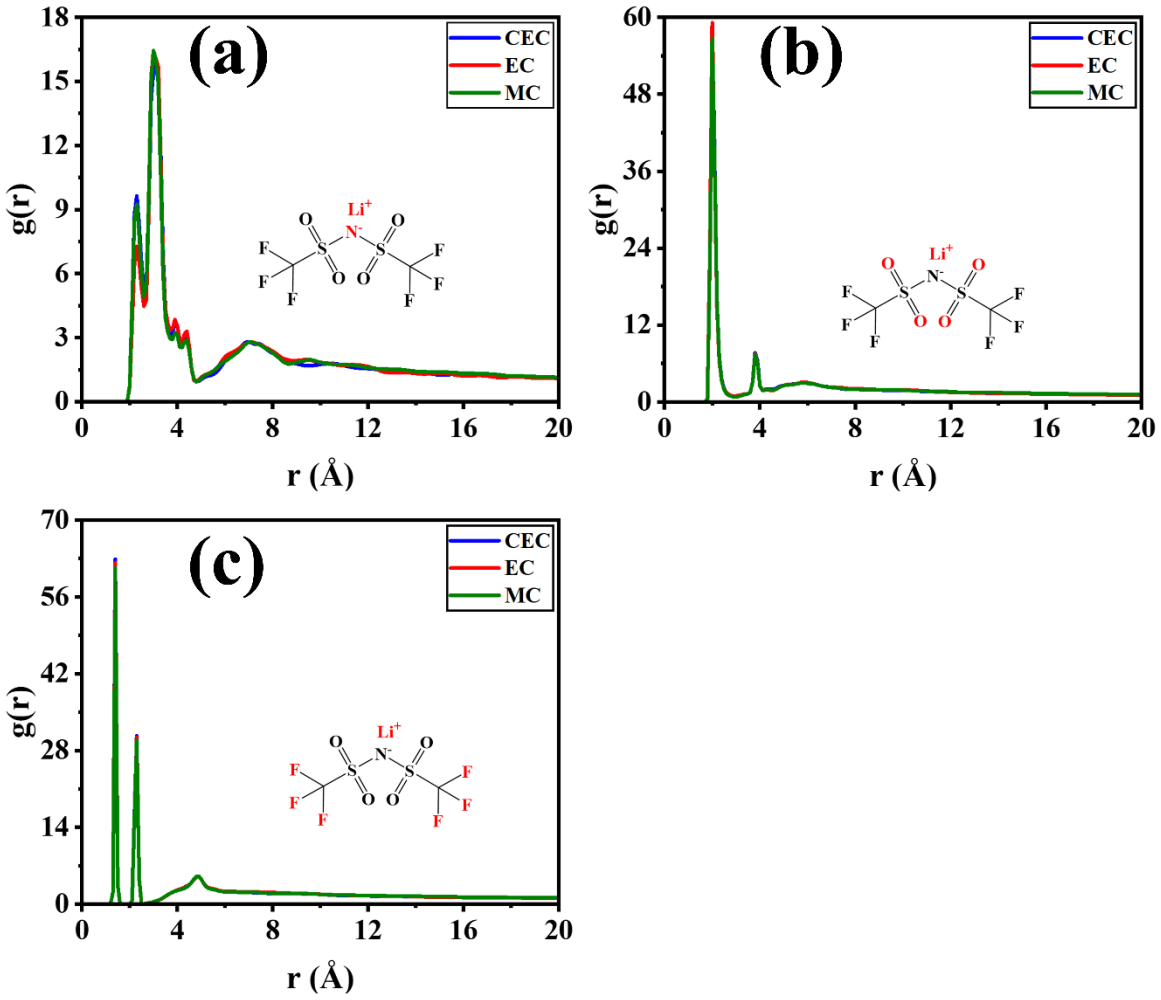


Figure S4. RDF of lithium-ion of LiTFSI to the other moiety of LiTFSI (a) nitrogen moiety of LiTFSI (b) oxygen moiety of LiTFSI (c) fluorine moiety of LiTFSI

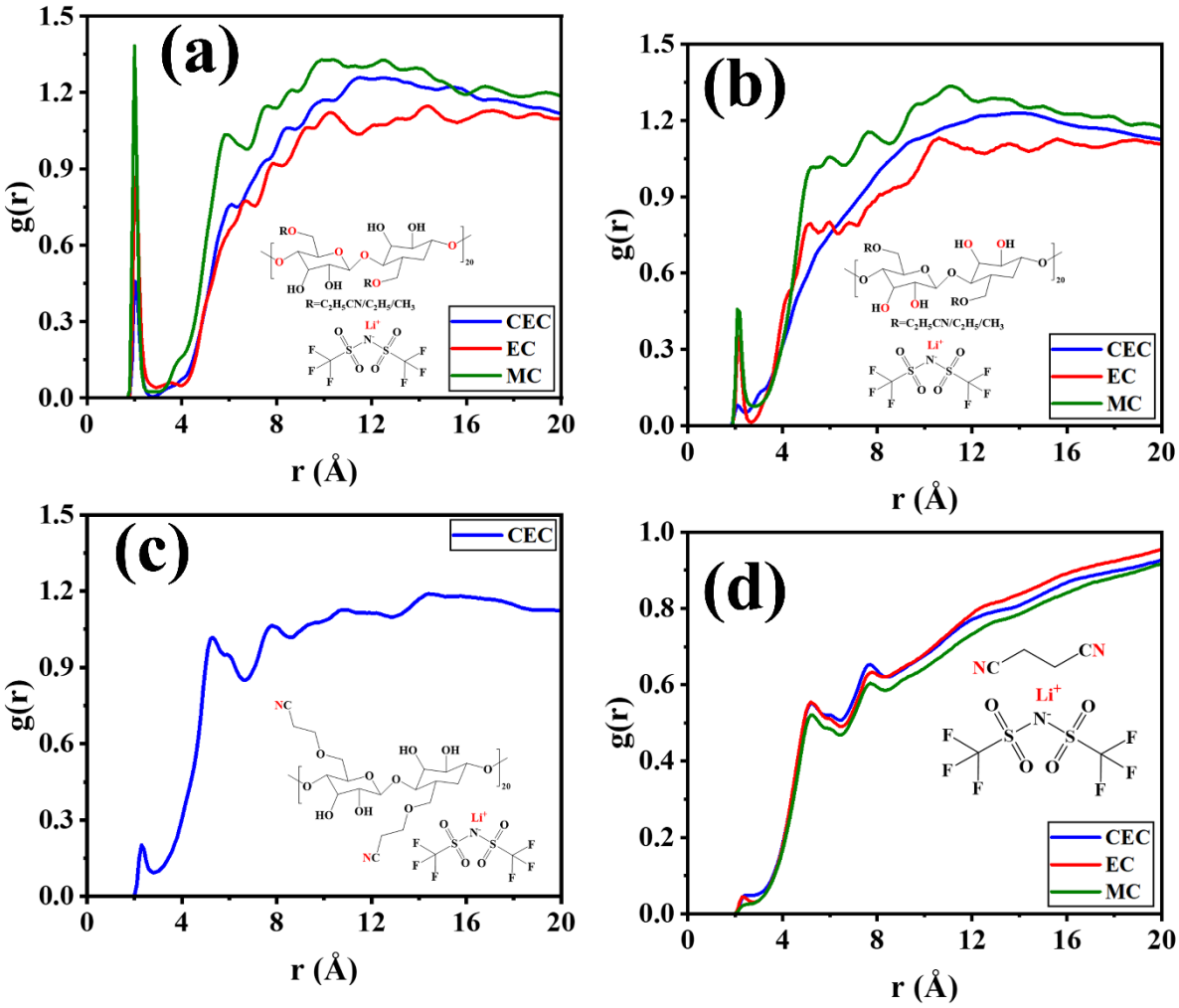


Figure S5. RDF of lithium-ion of LiTFSI to the other moiety of electrolytes (a) ether moiety of the polymer matrix (b) oxygen moiety of hydroxyl group presents in the polymer matrix (c) nitrogen moiety of the CEC system (d) nitrogen moiety present in the SN

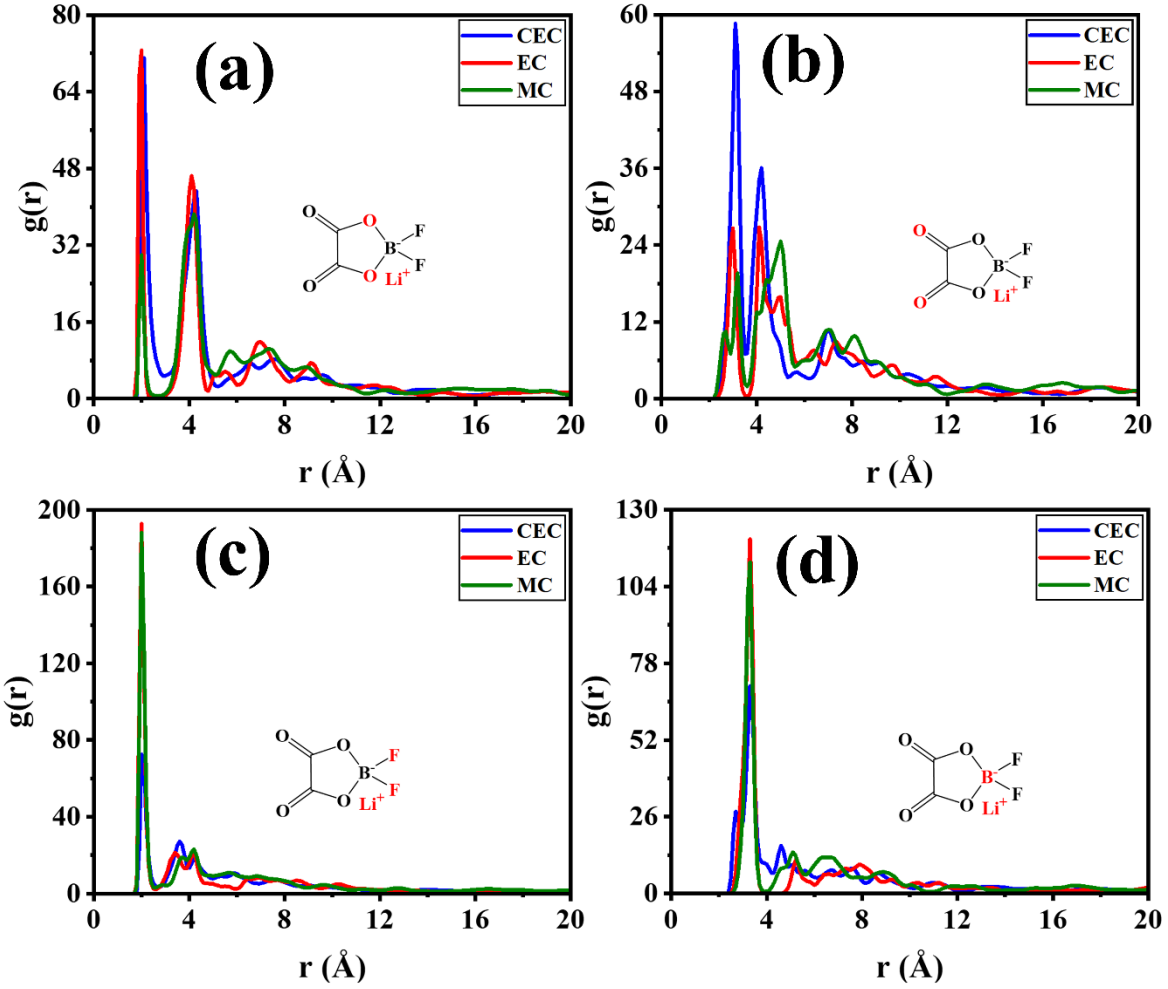


Figure S6. RDF between the lithium-ion of LiDFOB to the other moiety of LiDFOB (a) ether moiety of LiDFOB (b) keto moiety of LiDFOB (c) flurine moiety of LiDFOB (d) boron moiety of LiDFOB

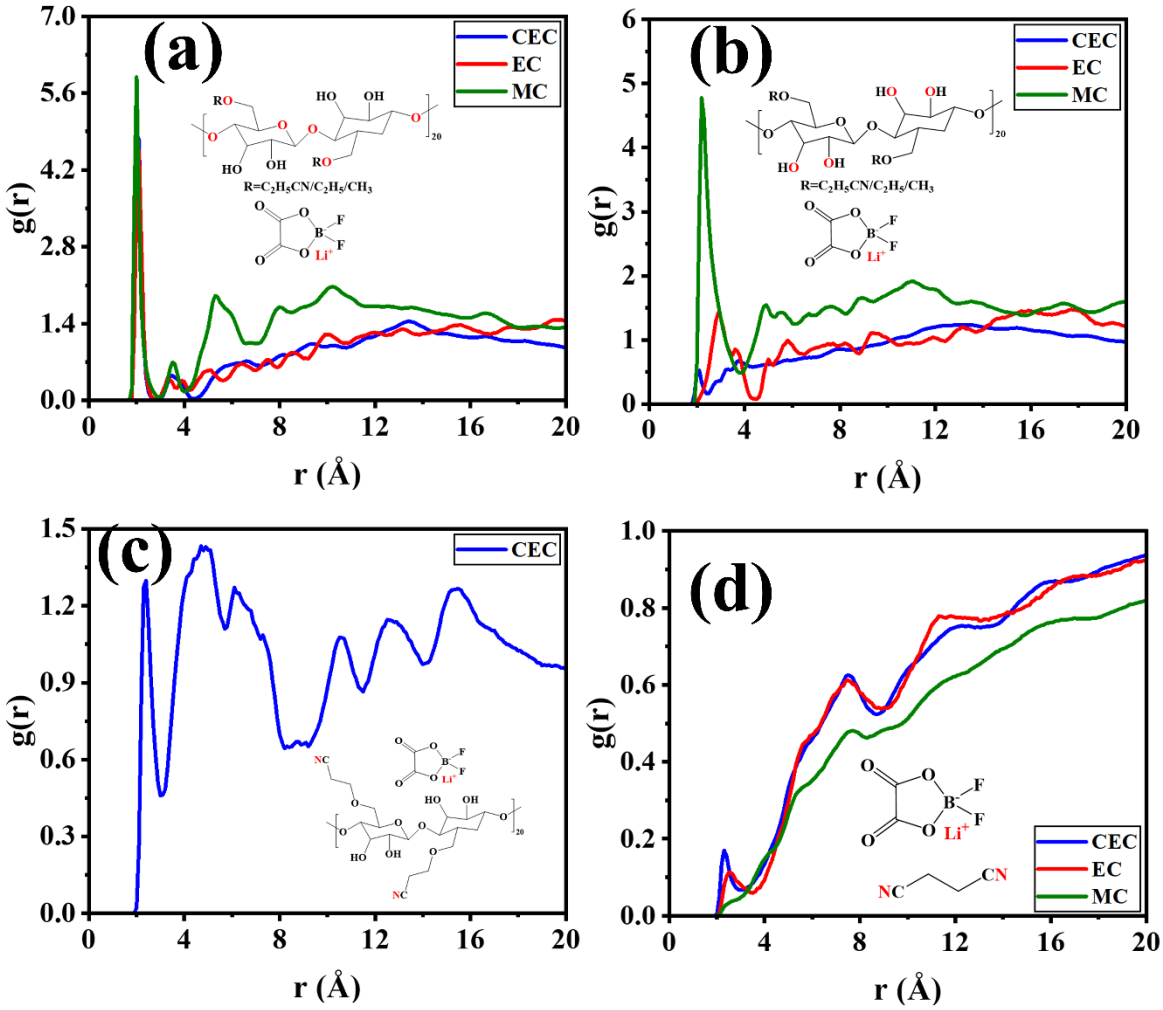


Figure S7. RDF of lithium-ion of LiDFOB to the other moiety of electrolytes (a) ether moiety of the polymer matrix (b) oxygen moiety of hydroxyl group presents in the polymer matrix (c) nitrogen moiety of the CEC system (d) nitrogen moiety present in the SN

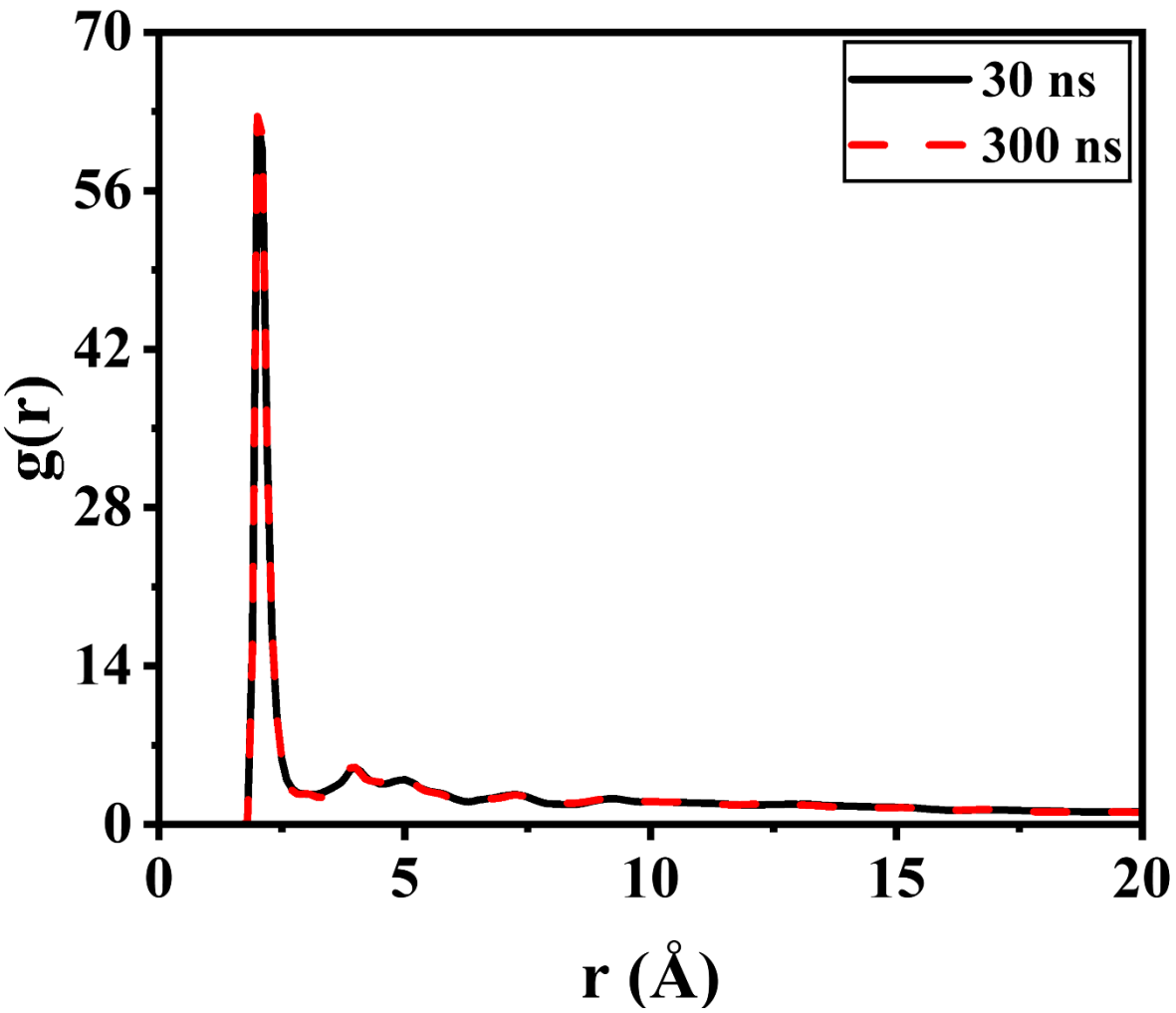


Figure S8. RDF of lithium-ions to the oxygen of the LiTFSI at the different time scales for the EC system

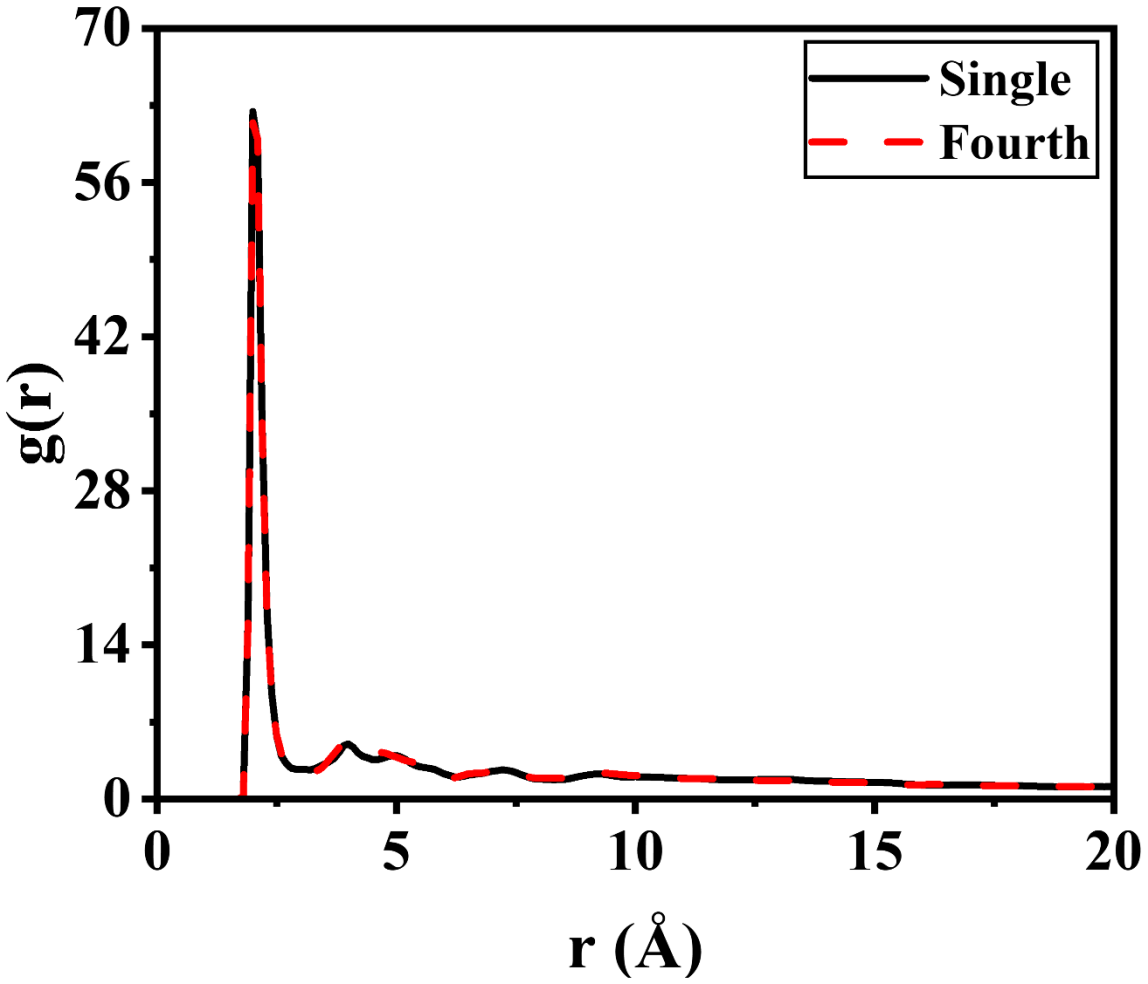


Figure S9. RDF of lithium-ions to the oxygen of the LiTFSI at different number of molecules for the EC system

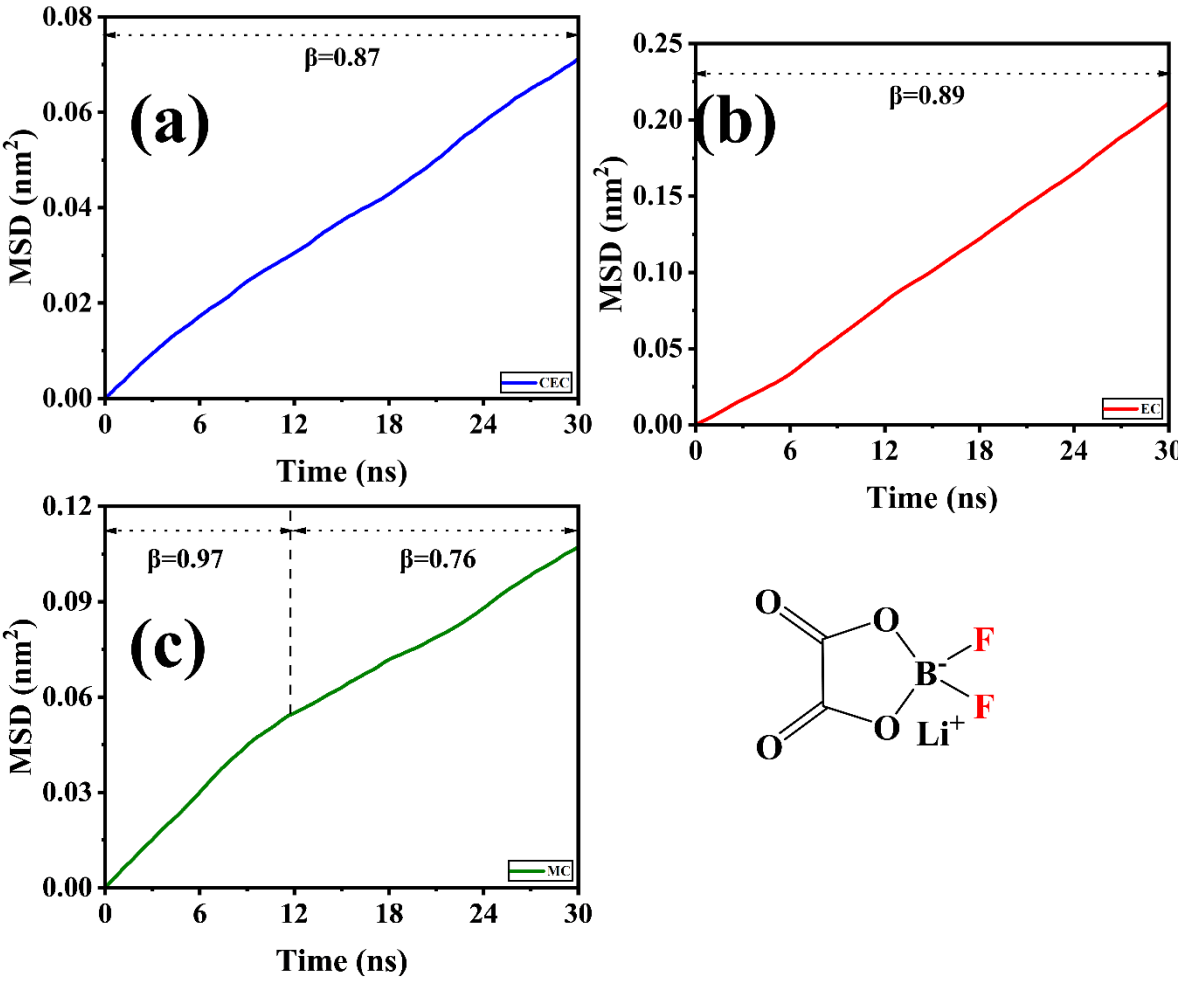


Figure S10. MSD of the fluorine moiety present in the LiDFOB (a) CEC (b) EC (c) MC

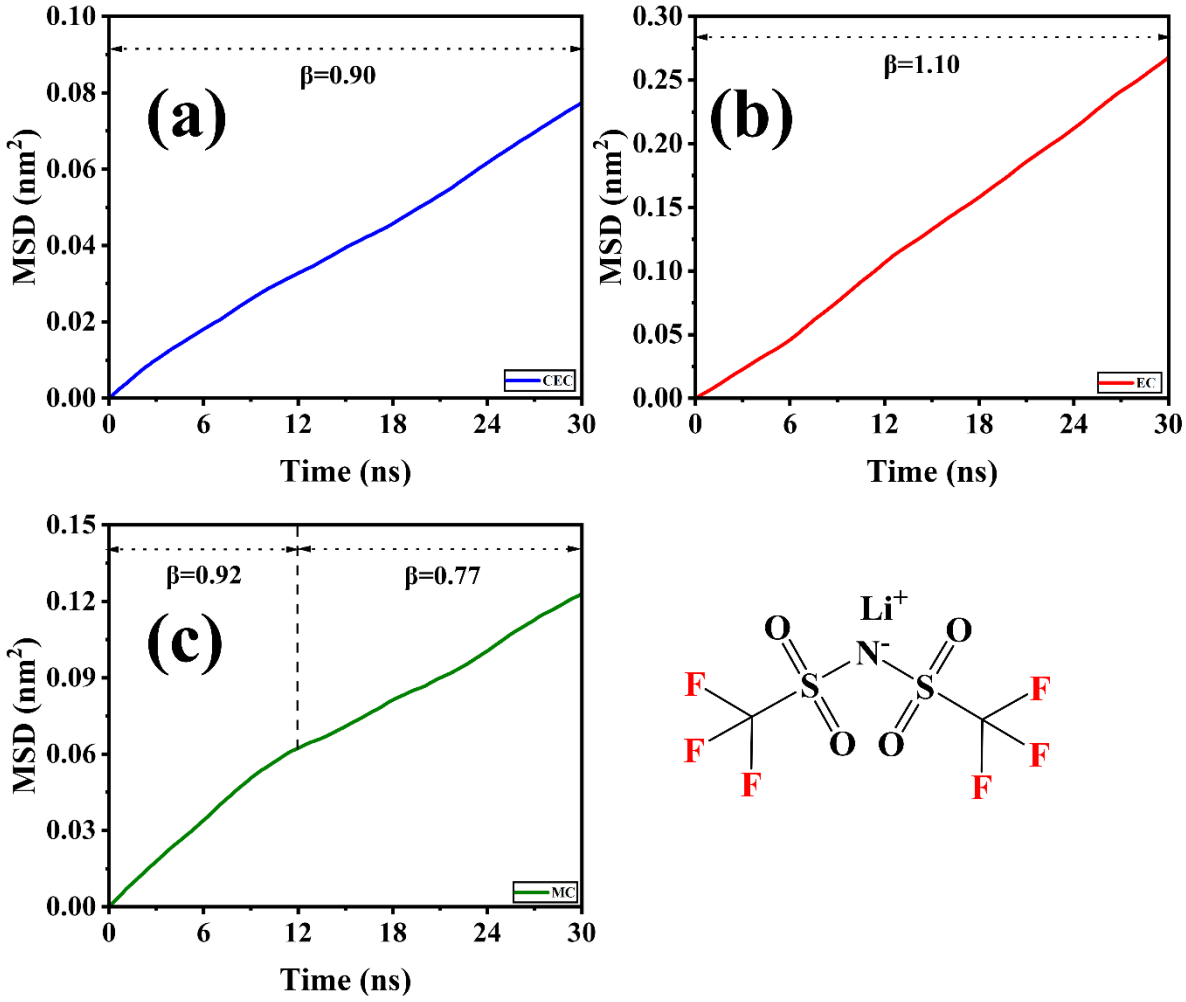


Figure S11. MSD of the fluorine moiety present in the LiTFSI (a) CEC (b) EC (c) MC

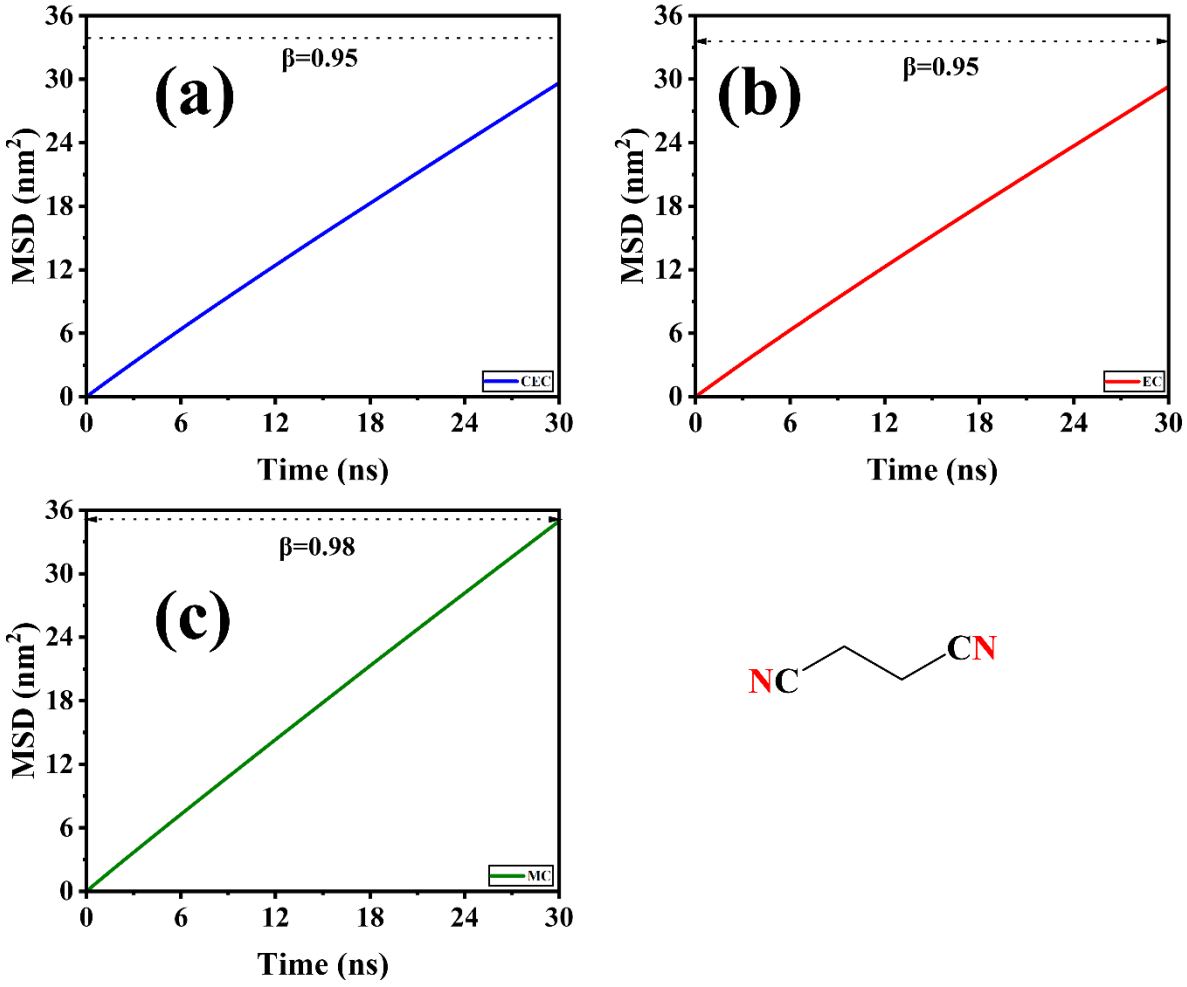


Figure S12. MSD of the nitrogen moiety present in the SN (a) CEC (b) EC (c) MC

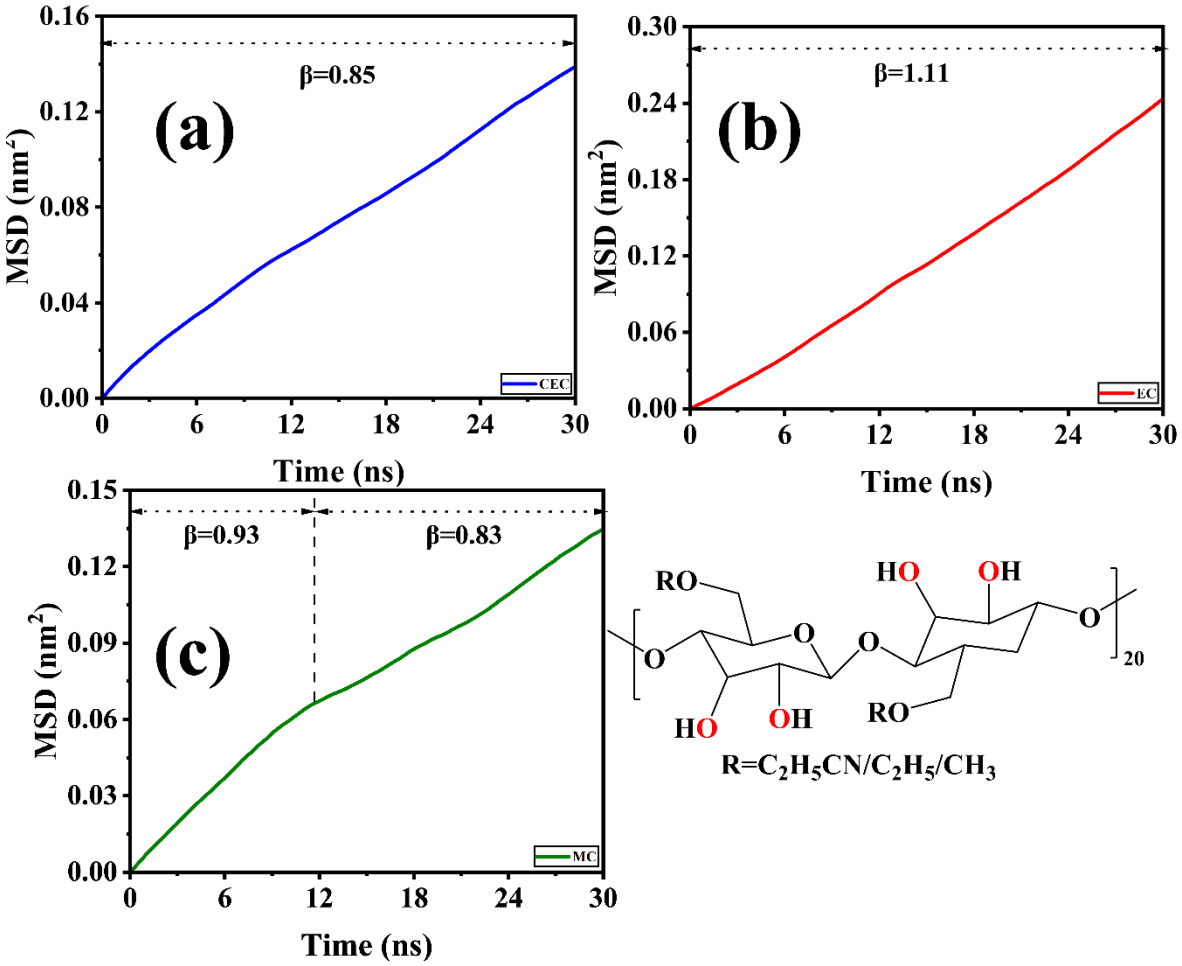


Figure S13. MSD of the oxygen moiety presents in the hydroxyl groups presents in the polymer matrix (a) CEC (b) EC (C) MC

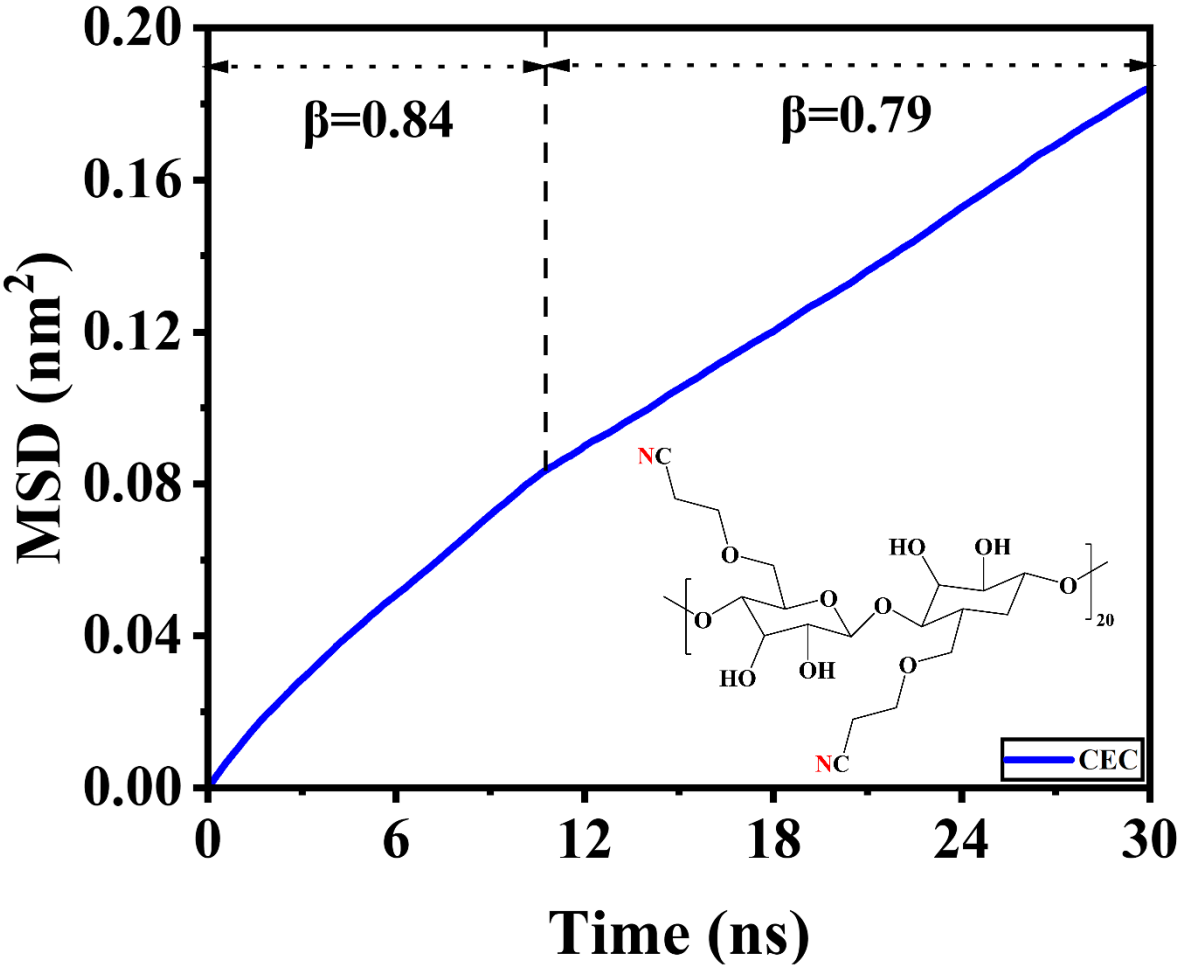


Figure S14. MSD of the nitrogen moiety present in the CEC polymer matrix